

Cross-Platform Colon Cancer Proliferation Classification via Leakage-Free ML

Calibration Benchmark | StabilitySelector (custom algorithm) | GDSC2 Drug Screen

Problem & Motivation

Colon cancer: 2nd leading cause of cancer death
Ki-67 IHC is subjective, single-gene, no drug prediction
Can ML predict proliferation class from transcriptomics?

CS Contribution

5 calibration methods x 4 model classes compared
StabilitySelector: bootstrap-based feature selection
Bonferroni-corrected drug screen (295 drugs)

Key Result

Cross-platform AUC = 0.973 (RF, no calibration)
Ki-67 correlation r = 0.59 (gene held out)
Trametinib p = 1.8e-12 (Bonferroni survives)
TCGA PanCancer survival p = 0.009

Pipeline & Custom Algorithm

GEO: GSE39582 (n=585) + GSE17538 (n=232)
Remove 10 proliferation genes (leakage prevention)
StabilitySelector: bootstrap 100x, keep stable features
Nested CV: 4 models (LR, RF, XGB, MLP)
TCGA-COAD (n=329) + CPTAC-COAD (n=105)
Calibration: 5 methods with bootstrap 95% CIs
GDSC2: 295 drugs x 969 lines (Bonferroni)

StabilitySelector: Novel sklearn-compatible transformer

Resamples data B=100 times, selects features in top K across >=50% of resamples. More robust than SelectKBest.

Results

Calibration Benchmark (TCGA)

RF Platt: AUC=0.973, ECE=0.04 (CIs sig. lower)
LR QN+Platt: AUC=0.972 (cross-platform)
XGB Platt: AUC=0.968, ECE=0.04

Ablation: StabilitySelector vs SelectKBest

LR: 0.994 both | RF: 0.988 vs 0.988
XGB: 0.990 vs 0.991 | MLP: 0.992 vs 0.983
-> StabilitySelector matches or beats SelectKBest

Drug Sensitivity (Bonferroni)

Trametinib p=1.8e-12 MEK
PD0325901 p=5.9e-12 MEK
SCH772984 p=1.1e-10 ERK
5/5 top hits = MAPK/ERK pathway

Survival

TCGA PanCancer log-rank p=0.009
GEO GSE39582 log-rank p=0.037